Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 1

be taken to imply a correct method.

Q	Working	Answer	Mark	Notes
1 (a)		e^6	1	B1 cao
(b)	$x^2-3x+x-3$		2	M1 for any 3 correct terms **or** for 4 out of 4 correct terms ignoring signs **or** for $x^2-2x\ldots$ **or** for $\ldots-2x-3$
	Correct answer scores full marks (unless from obvious incorrect working)	x^2-2x-3		A1
				Total 3 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 11

11	$\frac{9x}{12x}+\frac{2(5-x)}{12x}$ oe or or $\frac{3(6x)}{24x}+\frac{4(5-x)}{24x}$ oe or $\frac{3(6x)}{4(6x)}+\frac{4(5-x)}{4(6x)}$ oe or or $\frac{18x}{24x}+\frac{20-4x}{24x}$ oe or $\frac{3\times3x+2(5-x)}{12x}$ oe		3	M1 for two correct fractions with common denominator with the intention to add **or** a single correct fraction
	$\frac{9x+10-2x}{12x}$ oe or $\frac{18x+20-4x}{24x}$ oe or $\frac{14x+20}{24x}$ oe or			M1 for a correct numerator over a single denominator with brackets expanded and correct signs Allow $\frac{7x}{12x}+\frac{10}{12x}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$\frac{7x+10}{12x}$		A1 oe $\frac{10+7x}{12x}$
				Total 3 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 11

11		(definition of part: there are 3 parts: one part is the number, one part the letter t and one part the letter w Definition of terms: there are 6 terms: 2 number terms, 2 terms in t and 2 terms in w)		3	M1 indep	Fully correct cancellation of any two parts of their fraction at any stage of working
					M1 indep	correctly apply the negative power to the whole of their bracket (all parts or all terms) **or** correctly square all parts or terms of their bracket **or** correctly apply the negative power AND square of at least two parts (maybe 4 terms) of their bracket
			$4t^4w$		A1	Allow $(2t\ w)$ after the correct answer
		ALTERNATIVE				
			$4t^4w$	3	M2	2 correct terms (M1 for 1 correct term)
					A1	Allow $(2t\ w)$ after the correct answer
						Total 3 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2019 · Question 11

11	E.g. $\frac{10x}{6x}-\frac{3(x+2)}{6x}$ or $\frac{10x-3(x+2)}{6x}$ $\frac{10x-3x-6}{6x}$ **or** $\frac{7}{6x}-\frac{1}{x}$	$\frac{7x-6}{6x}$	3	M1	for two correct fractions with common denominator **or** a single correct fraction
				M1	for a correct single fraction with brackets expanded
				A1	for $\frac{7x-6}{6x}$ as the final answer SC: If no marks awarded then award B1 for an answer of $\frac{7x+6}{6x}$

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 12

12	eg $\frac{5(6x)}{24x}+\frac{4(x-3)}{24x}$ oe **or** $\frac{5(6x)}{4(6x)}+\frac{4(x-3)}{4(6x)}$ oe **or** $\frac{30x}{24x}+\frac{4(x-3)}{24x}$ oe **or** $\frac{30x+4(x-3)}{24x}$ oe **or** $\frac{15x}{12x}+\frac{2(x-3)}{12x}$ oe **or** $\frac{15x+2(x-3)}{12x}$ oe		3	M1	for two correct fractions with common denominator **or** a single correct fraction
	eg $\frac{30x+4x-12}{24x}$ oe **or** $\frac{30x}{24x}+\frac{4x-12}{24x}$ oe **or** $\frac{30x}{24x}+\frac{4x}{24x}-\frac{12}{24x}$ oe **or** $\frac{34x}{24x}-\frac{12}{24x}$ oe **or** $\frac{34x-12}{24x}$ oe **or** $\frac{15x+2x-6}{12x}$ oe			M1	for correct fraction(s) with bracket(s) expanded correctly
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{17x-6}{12x}$		A1	oe but must be simplified eg $\frac{-6+17x}{12x}$ do not ISW incorrect simplification eg $\frac{17x-6}{12x}=\frac{11}{12}$ is M2A0
					Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 12

Q	Working	Answer	Mark	Notes		
12	$3x(2x+5)=6x^2+15x$ or $3x(7x-4)=21x^2-12x$ or $(2x+5)(7x-4)=14x^2-8x+35x-20$ $(14x^2+27x-20)$		3	M1	An expansion with only one error. Do not award this mark for $6x+15x+21x-12x$ or $(6x+15x)(21x-12x)$	M2 for 3 (out of a maximum of 4) of $42x^3+105x^2-24x^2-60x$ (M1 for 2 correct out of a maximum of 4)
	$(6x^2+15x)(7x-4)=42x^3-24x^2+105x^2-60x$ $(21x^2-12x)(2x+5)=42x^3+105x^2-24x^2-60x$ $3x(14x^2-8x+35x-20)=42x^3-24x^2+105x^2-60x$ $3x(14x^2+27x-20)=42x^3+81x^2-60x$			M1	ft dep on M1 allow one further error	
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$42x^3+81x^2-60x$		A1	cao (terms may be in any order but must be simplified) dep on M1 ISW correct factorisation eg $3(14x^3+27x^2-20x)$ Do not ISW incorrect simplification eg $14x^3+27x^2-20x$	
						Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 13

13	$3x(2x-5)=6x^2-15x$ **or**		3	M1 for multiplying $3x$ by $(2x-5)$ with **both** terms correct or
	$(2x-5)^2=4x^2-10x-10x+25$ **or**			for multiplying $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(2x-5)^2=4x^2-20x+25$			for multiplying $(2x-5)$ by $(2x-5)$ and getting $4x^2-20x\ldots$ **or** $\ldots -20x+25$ (not for $4x^2+25$)
	$(6x^2-15x)(2x-5)=12x^3-30x^2-30x^2+75x$ oe **or**			M1ft (dep) for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(6x^2-15x)(2x-5)=12x^3-60x^2+75x$ oe **or**			for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ and getting $12x^3-60x^2\ldots$ **or** $\ldots -60x^2+75x$
	$3x(4x^2-10x-10x+25)=12x^3-30x^2-30x^2+75x$ oe **or**			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 3 out of 4 terms correct or
	$3x(4x^2-20x+25)=12x^3-60x^2+75x$			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 2 out of 3 terms correct or
				Expansion in one stage will lead to $12x^3-30x^2-30x^2+75x$ without firstly expanding two factors – award M2 for 3 out of 4 terms correct M1 for 2 out of 4 terms correct
	Working required	$12x^3-60x^2+75x$		A1 dep on M1
				Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 13

13		$4x(3x+1)=12x^2+4x$ **or** $4x(2x-3)=8x^2-12x$ **or** $(3x+1)(2x-3)=6x^2-9x+2x-3$ $(=6x^2-7x-3)$		3	M1 for expanding two of the three factors, allow one error
		$(12x^2+4x)(2x-3)=24x^3-36x^2+8x^2-12x$ oe $(8x^2-12x)(3x+1)=24x^3+8x^2-36x^2-12x$ oe $4x(6x^2-7x-3)=$ eg $24x^3-28x^2\ldots$. oe			M1 (dep)ft for expanding by the third factor, allow one error (some may do the expansion in one stage and will get to $24x^3-36x^2+8x^2-12x$ without firstly expanding two factors)
		Working required	$24x^3-28x^2-12x$		A1 dep on M1 isw correct factorisation eg $4(6x^3-7x^2-3x)$ $x(24x^2-28x-12)$ $4x(6x^2-7x-3)$ do not isw incorrect simplification eg $24x^3-28x^2-12x=6x^3-7x^2-3x$ gets M2A0
					Total 3 marks

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Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 14

Question	Working	Answer	Mark	Notes
14		$\frac{4y^5}{3x^2}$	3	B3 Accept $\frac{4}{3}x^{-2}y^5$ or $\frac{4x^{-2}y^5}{3}$ or $1.\dot{3}x^{-2}y^5$ oe NB: Must see 4 and 3 and not $16^{\frac{1}{2}}$ or $9^{\frac{1}{2}}$ or $16^{-\frac{1}{2}}$ or $9^{-\frac{1}{2}}$ (allow use of 1.3[33..])
				If not B3 then B2 for 2 of: correct fraction ($\frac{4}{3}$ or $1.\dot{3}$)(allow use of 1.3[33..]) **or** x term correct (x^2 on denominator or x^{-2} on numerator) **or** y term correct (y^5 on numerator or y^{-5} on denominator)
				If not B2 then **B1** for 1 of : correct fraction **or** x term correct **or** y term correct **or** for one of applying negative power to at least 3 out of 4 of $9, x^4, 16, y^{10}$ **or** applying square root to at least 3 out of 4 of $9, x^4, 16, y^{10}$ eg at least 3 of the 4 parts of $\frac{16y^{10}}{9x^4}$ **or** $\frac{16x^{-4}}{9y^{-10}}$ **or** $\frac{\frac{1}{9}x^{-4}}{\frac{1}{16}y^{-10}}$ **or** $\frac{3x^2}{4y^5}$ oe
				Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 14

14	$(4x+1)(x-3)=4x^2-12x+x-3(=4x^2-11x-3)$ $(4x+1)(5x+6)=20x^2+24x+5x+6(=20x^2+29x+6)$ $(x-3)(5x+6)=5x^2+6x-15x-18(=5x^2-9x-18)$		3	M1 for multiplying 2 brackets with at least 3 out of 4 terms correct
	$(5x+6)(4x^2-11x-3)=20x^3-55x^2-15x+24x^2-66x-18$ $(x-3)(20x^2+29x+6)=20x^3+29x^2+6x-60x^2-87x-18$ $(4x+1)(5x^2-9x-18)=20x^3-36x^2-72x+5x^2-9x-18$			M1 (dep) for multiplying the product of the first 2 brackets (ft from the 1st stage) by the 3rd bracket, and getting at least 3 out of 6 or 4 out of 8 terms correct
		$20x^3-31x^2-81x-18$		A1
	Alternative			
	$20x^3+24x^2-60x^2+5x^2-15x+6x-72x-18$			B2 for at least 6 out of 8 terms correct (B1 for 4 or 5 out of 8 correct terms)
		$20x^3-31x^2-81x-18$		A1
				Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 15

Question	Working	Answer	Mark	Notes
15		$125a^{12}c^6$	3	B3 for $125a^{12}c^6$ oe eg $\frac{125a^{12}}{c^{-6}}$ or $\frac{125c^6}{a^{-12}}$ or $\frac{125}{a^{-12}c^{-6}}$ or $\frac{a^{12}}{125^{-1}c^{-6}}$ or $\frac{1}{125^{-1}a^{-12}c^{-6}}$ or $\frac{a^{12}c^6}{125^{-1}}$ B2 for 2 correct terms (Allow eg $125a^{12}$ or $125c^6$ or $ka^{12}c^6$ as long as not added to any other terms) B1 for one correct term (allow eg a^{12} or $a^{-12}c^6$ or $\frac{1}{125^{-1}}$ or $\frac{1000a^{21}c^6}{8a^9}$ as long as not added to any other terms)
				Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 16

Question	Working	Answer	Mark	Notes
16	$(2x+3)(x-5)=2x^2-10x+3x-15(=2x^2-7x-15)$ or $(2x+3)(x+4)=2x^2+8x+3x+12(=2x^2+11x+12)$ or $(x-5)(x+4)=x^2+4x-5x-20(=x^2-x-20)$ oe		3	M1 For a correct method to expand two brackets with at least 3 terms correct out of 4 terms (or 2 terms correct out of 3 terms) Do not award this mark for eg $2x^2-10x+3x-15+x^2+4x-5x-20$ or eg $2x^2-10x+3x-15+x+4$
	$(2x^2-7x-15)(x+4)=2x^3+8x^2-7x^2-28x-15x-60$ or $(2x^2+11x+12)(x-5)=2x^3-10x^2+11x^2-55x+12x-60$ or $(x^2-x-20)(2x+3)=2x^3+3x^2-2x^2-3x-40x-60$ oe			M1ft Ft dep on M1 and a quadratic for a correct method to multiply by the 3rd bracket – allow one further error
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$2x^3+x^2-43x-60$		A1 If no working shown then award B2 for 3 out of a maximum of 4 terms correct
	ALTERNATIVE			
	$2x^3+8x^2-10x^2-40x+3x^2+12x-15x-60$		3	M2 For a complete expansion with 8 terms present of which 4 are correct (M1 for 4 correct terms from any number of terms)
		$2x^3+x^2-43x-60$		A1
				Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 18

Question		Working	Answer	Mark	Notes
18		$(2x-3)(2x+3)$ **or** $(2x+3)(x-4)$		3	M1 for correct factorisation of either the numerator **or** the denominator
		$(2x-3)(2x+3)$ **and** $(2x+3)(x-4)$			M1 for correct factorisation of the numerator **and** denominator
		Working required	$\frac{2x-3}{x-4}$		A1 (dep on M2) fully correct factorising and cancelling No ISW
					Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2019 · Question 22

22		$x(2x + 5)(3x - 1)$ or $(2x + 5)(3x^2 - x)$ or $(2x + 5)(2x - 5)$ oe			M1 for a correct factorisation of the numerator into 2 or 3 factors where one of the factors must be $(2x + 5)$ **or** denominator into 2 brackets where one of the factors must be $(2x + 5)$
		$x(2x + 5)(3x - 1)$ or $(2x + 5)(3x^2 - x)$ **and** $(2x + 5)(2x - 5)$ oe			M1 for a correct factorisation of the numerator into 2 or 3 factors where one of the factors must be $(2x + 5)$ **and** denominator into 2 brackets where one of the factors must be $(2x + 5)$
			$\frac{x(3x-1)}{2x-5}$	3	A1 accept $\frac{3x^2-x}{2x-5}$ oe Do not ISW
					Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 23

23	eg $\frac{20}{x^2-36}-\frac{2(x+6)}{x^2-36}$ oe **or** $\frac{20}{(x-6)(x+6)}-\frac{2(x+6)}{(x-6)(x+6)}$ oe **or** $\frac{20(x-6)}{(x^2-36)(x-6)}-\frac{2(x+6)(x-6)}{(x^2-36)(x-6)}$ or $\frac{20-2(x+6)}{(x^2-36)(4-x)}$ oe		3	M1	for writing the first two fractions with a common denominator (may be a single denominator) or multiplying both fractions by $\frac{1}{4-x}$ and writing over a common denominator
	eg $\frac{8-2x}{x^2-36}\times\frac{1}{4-x}$ **or** $\frac{8-2x}{(x-6)(x+6)}\times\frac{1}{4-x}$ **or** $\frac{20x-2x^2-48}{(x^2-36)(x-6)}\times\frac{1}{4-x}$ oe $\frac{8-2x}{(x^2-36)(4-x)}$ oe			M1	for simplifying first 2 fractions to a single fraction and expanding and simplifying numerator – must be correct, and showing intention to multiply by $\frac{1}{4-x}$ or expanding the numerator of the full solution and writing as a single fraction
		$\frac{2}{x^2-36}$		A1	oe eg $\frac{2}{(x-6)(x+6)}$
					Total 3 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 21

21		$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)}\times\frac{(x-5)(x-3)}{5x+8}$ or eg $\frac{(5x-8)(x-5)}{(5x+2)}(-(x-7))$		4	M2 For factorising at least 2 of the quadratics correctly – could be implied by 2 factors cancelled correctly (M1 For factorising at least 1 of the 3 quadratics correctly)
		$\frac{(5x-8)(x-5)-(x-7)(5x+2)}{5x+2}$ oe or $\frac{5x^2-25x-8x+40-(5x^2-35x+2x-14)}{5x+2}$ oe or $\frac{(25x^2-64)(x^2-8x+15)-(x-7)(5x^2-13x-6)(5x+8)}{(5x^2-13x-6)(5x+8)}$ oe or $\frac{(5x-8)(x^2-8x+15)-(x-7)(5x+2)(x-3)}{(5x+2)(x-3)}$ oe or $\frac{(25x^2-64)(x-5)-(x-7)(5x+2)(5x+8)}{(5x+2)(5x+8)}$ oe			M1 (indep (ft if M2 awarded)) For writing the fractions over a common denominator with or without brackets removed – need not be in simplest form Could be written as 2 separate fractions over a common denominator
			$\frac{54}{5x+2}$		A1 dep on M3
					Total 4 marks

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Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 23

23	$16x^2-36=(4x-6)(4x+6)$ [$=4(2x+3)(2x-3)$] oe or $16x^2-36=(8x-12)(2x+3)$ [$=4(2x+3)(2x-3)$] oe		4	M1indep	
	$2x^2+7x+6=(2x+3)(x+2)$ **and** $x^2-5x-14=(x-7)(x+2)$ [We will make an exception of $2x^2+7x+6=(4x+6)(0.5x+1)$ as this then cancels with $(4x+6)$]			M1indep	[NB: the two fractions when divided and cancelled give an answer of $4(2x-3)$ or $2(4x-6)$ or $8x-12$ (any one of these gain 2 marks)] M2 for any fraction with completely simplified non-linear numerator and non-linear denominator that will cancel to -19
	$4(2x-3)-(7+8x)(=n)$ [allow invisible brackets ie $8x-12-7+8x$] or an expression that clearly shows the numerator is -19 times the denominator eg $\frac{-38x-57}{2x+3}(=n)$			M1	a linear expression that should give the correct value for n (this mark implies previous M marks as not all factorising is necessary)
	Working required	-19		A1	dep on M2
					Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 24

24	$\frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)}$ or $\frac{8x-12-6x+15}{(2x-5)(2x-3)}$ oe		4	M1	Writing 1st fraction as a fraction over a common denominator (can be 2 separate fractions)
	$x(3-2x)(3+2x)$ or $(3x-1)(2x-5)$			M1	Complete factorisation of numerator or denominator of 2nd fraction
	$\frac{2x+3}{(2x-5)(2x-3)}\times\frac{(3x-1)(2x-5)}{x(3-2x)(3+2x)}$			M1	may be partially simplified
		$\frac{3x-1}{x(2x-3)(3-2x)}$		A1	e.g. $\frac{3x-1}{x(2x-3)(3-2x)}$ or $\frac{1-3x}{x(2x-3)^2}$ or $\frac{3x-1}{x(12x-9-4x^2)}$ or $\frac{3x-1}{(12x^2-9x-4x^3)}$ oe isw for incorrect denominator expansion
					Total 4 marks

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Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 26

26	$(3x-5)(x+2)$	$\frac{4(3x^2+x-10)-(3x-5)(4x-1)}{3x^2+x-10}$ oe or $\frac{27x-45}{3x^2+x-10}$		4	M1	for correctly factorising $3x^2+x-10$ to give $(3x-5)(x+2)$ (may be seen later on in working) **OR** combining 2 fractions into a correct single fraction
	$\frac{4x-1}{x+2}$ implies first M1	$\frac{4(3x^2+x-10)-(3x-5)(4x-1)}{(3x-5)(x+2)}$ or $\frac{27x-45}{(3x-5)(x+2)}$			M1	for inverting and cancelling giving a correct fraction **OR** for a correct single fraction where the denominator is factorised
	$\frac{4(x+2)-(4x-1)}{x+2}$ or $\frac{4x+8-4x+1}{x+2}$ or $\frac{4(x+2)}{x+2}-\frac{4x-1}{x+2}$ or $\frac{4x+8}{x+2}-\frac{4x-1}{x+2}$	$\frac{9(3x-5)}{(3x-5)(x+2)}$			M1	for a correct single fraction or two correct fractions with a common denominator **OR** for a correct fully factorised single fraction
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)		$\frac{9}{x+2}$		A1	
						Total 4 marks

CHALLENGE QUESTIONS

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 5

5	(a)		m^7	1	B1	
	(b)		c^8	1	B1	
	(c)		a^{15}	1	B1	
	(d)	$8x + 12 + 2x + 10$	$10x + 22$	2	M1 A1	Any three terms correct out of four. Allow $2(5x + 11)$ Do not ISW
						Total 5 marks

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Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 23

23	$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)} \ (=p)$ or $\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} \ (=p)$ or 2 from $4(x-6)(x+5)$ or $5(x-6)(x+6)$ or $x(x+5)$ or $10x(x+6)$		4	M1	for factorising 2 or 3 of the quadratics fully – could be implied by 2 factors cancelled correctly NB factors must be in the form $(ax+b)$ **NB Substitution of values of x into the given equation is not an acceptable algebraic method**
	$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)} \ (=p)$ or $\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} \ (=p)$ or $4(x-6)(x+5)$ and $5(x-6)(x+6)$ and $x(x+5)$ and $10x(x+6)$			M1	for factorising **all** of the quadratics fully – could be implied by 2 factors cancelled correctly NB factors must be in the form $(ax+b)$
	$\frac{4x^2-4x-120}{5x^2-180} \times \frac{10x^2+60x}{x^2+5x} \ (=p)$			M1	for inverting the 2nd fraction (this mark can be awarded at any time and may be awarded with incorrect factorisation if meaning is clear)
	Working required	8		A1	oe dep on M3
					Total 4 marks
ALT	$\frac{40x^4+200x^3-1440x^2-7200x}{5x^4+25x^3-180x^2-900x}$			M3	for a correct expression (no errors)
		8		A1	oe dep on M3

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 23

23	$(x+2)(x-2)$ oe or $(4x+1)(x-2)$ oe		4	M1	for complete factorisation of x^2-4 **or** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$ where a and b are integers
	$(x+2)(x-2)\times\frac{x}{(4x+1)(x-2)}$ or $\frac{x(x+2)(x-2)}{(4x+1)(x-2)}$ or $\frac{x(x+2)}{(4x+1)}$			M1	for complete factorisation of $4x^2-7x-2$ **and** x^2-4 and inverting and intention to multiply
	$\frac{x(x+2)-2x(4x+1)}{(4x+1)}$ or $\frac{x^2+2x-8x^2-2x}{(4x+1)}$ or $\frac{x(x+2)}{(4x+1)}-\frac{2x(4x+1)}{(4x+1)}$ or $\frac{x^2+2x}{(4x+1)}-\frac{8x^2+2x}{(4x+1)}$			M1	for a correct single fraction following correct cancellation or for two correct fractions with common denominator following correct cancellation
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{-7x^2}{4x+1}$		A1	oe but must be in form $\frac{ax^2}{bx+c}$ where a, b and c are integers.
					Total 4 marks

23 **ALT**	$\frac{-7x^3+14x^2}{4x^2-7x-2}$ oe		4	M1	for a correct single fraction
	$\frac{-7x^2(x-2)}{(4x+1)(x-2)}$ oe			M1	for complete factorisation of $-7x^3+14x^2$ **or** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$
	$\frac{-7x^2(x-2)}{(4x+1)(x-2)}$ oe			M1	for complete factorisation of $-7x^3+14x^2$ **and** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{-7x^2}{4x+1}$		A1	oe but must be in form $\frac{ax^2}{bx+c}$ where a, b and c are integers.
					Total 4 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 26

26	eg $\frac{(4x+3)(x-5)}{2x-1}\times\frac{(2x-1)(x-3)}{(x+5)(x-5)}$ **or** eg $\frac{(4x+3)(x-3)}{x+5}(+(29-4x))$		4	M2	for factorising at least 2 of the quadratics correctly – could be implied by 2 factors cancelled correctly (M1 for factorising at least 1 of the 3 quadratics correctly)
	eg $\frac{(4x+3)(x-3)+(29-4x)(x+5)}{x+5}$ oe **or** eg $\frac{4x^2-9x-9+145+9x-4x^2}{x+5}$ oe			M1	for writing the correct fractions over a common denominator of $(x+5)$ with or without brackets removed – need not be in simplest form. Could be written as 2 separate fractions.
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{136}{x+5}$		A1	
					Total 4 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 21

21	$\left(\frac{9x^2-4}{3x^2-13x-10}=\right)\ \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1	for either $(3x+2)(3x-2)$ **or** $(3x+2)(x-5)$
	$\left(\frac{9x^2-4}{3x^2-13x-10}=\right)\ \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1	for $(3x+2)(3x-2)$ **and** $(3x+2)(x-5)$
					M2 for $\frac{9x^2-4}{(9x^2-4)(x-5)}=\frac{1}{(x-5)}$
	E.g. of denominators $(3x-2)(3x^2-13x-10)(x-1)$ **or** $(3x-2)(3x+2)(x-5)(x-1)$ **or** $9x^4-54x^3+41x^2+24x-20$ **or** $(3x+2)(x-5)(x-1)$ **or** $3x^3-16x^2+3x+10$ **or** $(3x-2)(x-5)(x-1)$ **or** $3x^3-20x^2+27x-10$ **or** $(x-5)(x-1)$ **or** x^2-6x+5			M1	(indep) ft their fractions for use of a correct common denominator for 2 fractions with **algebraic** denominators NB: fractions need not be simplified
	$\frac{x-1-7(x-5)}{(x-5)(x-1)}$ **or** $\frac{x-1-7x+35}{(x-5)(x-1)}$ **or** $\frac{x-1-7(x-5)}{x^2-6x+5}$ **or** $\frac{x-1-7x+35}{x^2-6x+5}$ oe			M1	for a **correct** fraction with a **correct quadratic** denominator – may or may not be expanded which leads to a correct answer
		$\frac{2(17-3x)}{(x-5)(x-1)}$	5	A1	accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct
					Total 5 marks

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